

Md Badrul Hasan

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Physics-Based Simulation | ML Surrogate Modeling | CFD & Heat Transfer |
Scientific Python & HPC

Professional Summary

Postdoctoral researcher with a Ph.D. in Mechanical Engineering and 5+ years building physics-based simulations and machine-learning surrogate models for coupled fluid flow, heat transfer, and species transport. Experienced designing neural network surrogates that replace expensive simulation computations, validating models against high-fidelity reference data, and deploying learned models into production solvers on HPC clusters. Skilled in ANSYS Fluent, COMSOL Multiphysics, and custom finite-volume solvers, with strong communication skills for translating complex technical work to multidisciplinary teams.

Technical Skills

- **CFD & Simulation Software:** ANSYS Fluent (including UDFs), COMSOL Multiphysics, OpenFOAM (familiarity), custom finite-volume and finite-difference solvers in Fortran and Python
- **Analysis Domains:** Coupled heat transfer and fluid flow, species transport and contaminant dispersion, turbulence modeling (RANS, LES)
- **Programming:** Python (NumPy, SciPy, pandas, matplotlib), MATLAB, Fortran, Bash, C++ (familiarity); Git/GitHub version control
- **Machine Learning & Surrogate Modeling:** PyTorch, TensorFlow, scikit-learn; physics-constrained neural networks, surrogate models for expensive simulations, model validation against experimental and high-fidelity reference data
- **Numerical Methods:** Finite-difference, finite-volume, and spectral methods; solver verification and validation; grid-convergence studies; numerical stability and dissipation analysis
- **HPC & Workflow Automation:** MPI/OpenMPI parallel computing, Slurm scheduling, Linux clusters, automated batch workflows for large-scale simulations
- **Communication:** Technical report and white paper writing, client-facing presentations, cross-discipline coordination

Research Experience

Postdoctoral Researcher, Harold Vance Department of Petroleum Engineering *Oct 2026 to Present*
Texas A&M University, ONR-funded project with Deltares (PI: Dr. Wencheng Jin) *College Station, TX*

- Accelerating the Fortran-based XBeach coastal storm-impact solver on GPUs with nvFortran and OpenACC; profiled a 72-hour Hurricane Sandy hindcast (84,537 cells) to find the wave-action solver as the main bottleneck (53–86% of runtime) and measured $51\times$ strong scaling at 143 MPI ranks.
- Verified solver correctness across gfortran and nvFortran builds (bitwise round-off agreement on final bed level); developing JAX-based ML closures for wave hydrodynamics and a neural emulator for morphodynamics.

Graduate Research Assistant, Computational Mechanics Laboratory, UMBC *Jan 2022 to Aug 2026*
Department of Mechanical Engineering, University of Maryland Baltimore County *Baltimore, MD*

- Designed neural network surrogate models to replace expensive physics computations in production CFD solvers, embedding physical constraints (symmetry, invariance, realizability) directly in model architecture, demonstrated computational speedup while maintaining physical accuracy.
- Developed and validated CFD models of turbulent flows with coupled heat transfer, fluid dynamics, and species transport; benchmarked predictions against reference data across the grid resolutions on HPC clusters.
- Built automated Python pipelines to process terabyte-scale simulation outputs; performed grid-convergence and accuracy studies to validate subgrid-scale model predictions against high-fidelity reference data.

Graduate Research Assistant, Joint Center for Earth Systems Technology (JCET) *Jan 2020 to Jun 2022*
UMBC, in collaboration with NOAA-affiliated research teams *Baltimore, MD*

- Quantified how discretization choices affect energy and momentum transport fidelity in PDE-based solvers; benchmarked accuracy across grid resolutions and identified scheme-dependent error sources to guide solver selection.
- Processed and analyzed airborne observational data; correlated measurements with simulation outputs to validate model predictions under extreme flow conditions.

Key Research Projects

CFD Surrogate Modeling for Transport Applications *2023 to Present*

- Designed neural network surrogates to replace expensive subgrid-scale stress computations, embedding physical constraints directly in model architecture (arXiv:2504.14473, under review).
- Integrated learned closures into large-scale CFD solvers; evaluated numerical stability, physical accuracy, and computational speedup through systematic a priori and a posteriori testing.

Education

University of Maryland, Baltimore County (UMBC) *Baltimore, MD*
Ph.D., Mechanical Engineering *Advisor: Dr. Meilin Yu* *June 2026*
Dissertation: Invariance-Embedded ML Subgrid-Scale Models for Hurricane Boundary Layer Simulations: Development and A priori/A posteriori Tests with WRF

University of Maryland, Baltimore County (UMBC) *Baltimore, MD*
M.S., Mechanical Engineering *Advisor: Dr. Meilin Yu, Dr. Stephen Guimond* *Dec 2022*
Thesis: Numerical Dissipation Effects on Mesoscale Hurricane Intensification Simulations

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*
B.Sc., Mechanical Engineering *2017*